

Package ‘Strategus’

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Type Package

Title Coordinate and Execute OHDSI HADES Modules

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Description Coordinate and execute large scale analytics using OHDSI Health Analytics Data-to-Evidence Suite (HADES) (<<https://ohdsi.github.io/Hades/>>) modules.

Depends R (>= 4.2.0),
CohortGenerator (>= 0.11.0),
DatabaseConnector (>= 6.2.3),
R6

Imports checkmate,
cli,
digest,
dplyr,
methods,
ParallelLogger (>= 3.1.0),
purrr,
ResultModelManager (>= 0.5.8),
rlang,
SqlRender (>= 1.18.0)

Suggests Characterization,
CirceR,
CohortDiagnostics (>= 3.4.0),
CohortIncidence,
CohortMethod,
Cyclops (>= 3.6.0),
Eunomia,
EvidenceSynthesis (>= 1.0.0),
FeatureExtraction,
fs,
knitr,
PatientLevelPrediction (>= 6.5.0),
readr,
rmarkdown,
RSQLite,
SelfControlledCaseSeries (>= 6.0.0),

testthat (>= 3.0.0),
 TreatmentPatterns (>= 3.1.0),
 withr

Remotes ohdsi/CohortDiagnostics,
 ohdsi/CohortIncidence,
 ohdsi/CohortMethod

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VignetteBuilder knitr

URL <https://ohdsi.github.io/Strategus>, <https://github.com/OHDSI/Strategus>

BugReports <https://github.com/OHDSI/Strategus/issues>

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Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications\(\)](#)

moduleSpecifications

Created by the [CohortDiagnosticsModule\\$createModuleSpecifications\(\)](#) function.

Value

Returns the analysisSpecifications object with the module specifications added.

addCohortGeneratorModuleSpecifications

Add Cohort Generator module specifications to analysis specifications

Description

Add Cohort Generator module specifications to analysis specifications

Usage

```
addCohortGeneratorModuleSpecifications(
  analysisSpecifications,
  moduleSpecifications
)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications\(\)](#)

moduleSpecifications

Created by the [CohortGeneratorModule\\$createModuleSpecifications\(\)](#) function.

Value

Returns the analysisSpecifications object with the module specifications added.

addCohortIncidenceModuleSpecifications

Add Cohort Incidence module specifications to analysis specifications

Description

Add Cohort Incidence module specifications to analysis specifications

Usage

```
addCohortIncidenceModuleSpecifications(  
  analysisSpecifications,  
  moduleSpecifications  
)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications](#)

moduleSpecifications

Created by the [CohortIncidenceModule\\$createModuleSpecifications\(\)](#) function.

Value

Returns the analysisSpecifications object with the module specifications added.

addCohortMethodeModuleSpecifications

Add Cohort Method module specifications to analysis specifications

Description

Add Cohort Method module specifications to analysis specifications

Usage

```
addCohortMethodeModuleSpecifications(  
  analysisSpecifications,  
  moduleSpecifications  
)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications](#)

moduleSpecifications

Created by the [CohortMethodModule\\$createModuleSpecifications\(\)](#) function.

Value

Returns the analysisSpecifications object with the module specifications added.

```
addEvidenceSynthesisModuleSpecifications
```

Add Evidence Synthesis module specifications to analysis specifications

Description

Add Evidence Synthesis module specifications to analysis specifications

Usage

```
addEvidenceSynthesisModuleSpecifications(
  analysisSpecifications,
  moduleSpecifications
)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications](#)

moduleSpecifications

Created by the [EvidenceSynthesisModule\\$createModuleSpecifications\(\)](#) function.

Value

Returns the analysisSpecifications object with the module specifications added.

```
addModuleSpecifications
```

Add generic module specifications to analysis specifications

Description

Add generic module specifications to analysis specifications

Usage

```
addModuleSpecifications(analysisSpecifications, moduleSpecifications)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications](#)

moduleSpecifications

An object of type ModuleSpecifications

Value

Returns the analysisSpecifications object with the module specifications added.

```
addPatientLevelPredictionModuleSpecifications
```

Add Patient Level Prediction module specifications to analysis specifications

Description

Add Patient Level Prediction module specifications to analysis specifications

Usage

```
addPatientLevelPredictionModuleSpecifications(  
  analysisSpecifications,  
  moduleSpecifications  
)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications](#)

moduleSpecifications

Created by the [PatientLevelPredictionModule\\$createModuleSpecifications\(\)](#) function.

Value

Returns the analysisSpecifications object with the module specifications added.

```
addPatientLevelPredictionValidationModuleSpecifications
```

Add Patient Level Prediction Validation Module module specifications to analysis specifications

Description

Add Patient Level Prediction Validation Module module specifications to analysis specifications

Usage

```
addPatientLevelPredictionValidationModuleSpecifications(  
  analysisSpecifications,  
  moduleSpecifications  
)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications](#)

moduleSpecifications

Created by the [PatientLevelPredictionValidationModule\\$createModuleSpecifications\(\)](#) function.

Value

Returns the analysisSpecifications object with the module specifications added.

addSelfControlledCaseSeriesModuleSpecifications

Add Self Controlled Case Series Module module specifications to analysis specifications

Description

Add Self Controlled Case Series Module module specifications to analysis specifications

Usage

```
addSelfControlledCaseSeriesModuleSpecifications(
  analysisSpecifications,
  moduleSpecifications
)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications](#)

moduleSpecifications

Created by the [SelfControlledCaseSeriesModule\\$createModuleSpecifications\(\)](#) function.

Value

Returns the analysisSpecifications object with the module specifications added.

addSharedResources

Add shared resources (i.e. cohorts) to analysis specifications

Description

Add shared resources (i.e. cohorts) to analysis specifications

Usage

```
addSharedResources(analysisSpecifications, sharedResources)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications](#)

sharedResources

An object of type SharedResources.

Value

Returns the analysisSpecifications object with the module specifications added.

```
addTreatmentPatternsModuleSpecifications
```

Add Treatment Patterns Module specifications to analysis specifications

Description

Add Treatment Patterns Module specifications to analysis specifications

Usage

```
addTreatmentPatternsModuleSpecifications(
  analysisSpecifications,
  moduleSpecifications
)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications\(\)](#)

moduleSpecifications

Created by the `TreatmentPatternsModule$createModuleSpecifications()`

Value

Returns the analysisSpecifications object with the module specifications added

```
CharacterizationModule
```

Characterize cohorts with the R [href=https://ohdsi.github.io/Characterization/HADES](https://ohdsi.github.io/Characterization/HADES) Characterization Package

Description

Computes cohort characterization information against the OMOP Common Data Model

Super class

[Strategus::StrategusModule](#) -> CharacterizationModule

Public fields

tablePrefix The table prefix to append to the results tables

Methods

Public methods:

- [CharacterizationModule\\$new\(\)](#)
- [CharacterizationModule\\$execute\(\)](#)
- [CharacterizationModule\\$createResultsDataModel\(\)](#)
- [CharacterizationModule\\$getResultsDataModelSpecification\(\)](#)
- [CharacterizationModule\\$uploadResults\(\)](#)
- [CharacterizationModule\\$createModuleSpecifications\(\)](#)
- [CharacterizationModule\\$clone\(\)](#)

Method `new()`: Initialize the module

Usage:

```
CharacterizationModule$new()
```

Method `execute()`: Execute characterization

Usage:

```
CharacterizationModule$execute(
  connectionDetails,
  analysisSpecifications,
  executionSettings
)
```

Arguments:

`connectionDetails` An object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpecifications\(\)](#)

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpecifications\(\)](#)

`executionSettings` An object of type `ExecutionSettings` as created by [createCdmExecutionSettings\(\)](#) or [createResultsExecutionSettings\(\)](#).

Method `createResultsDataModel()`: Create the results data model for the module

Usage:

```
CharacterizationModule$createResultsDataModel(
  resultsConnectionDetails,
  resultsDatabaseSchema,
  tablePrefix = self$tablePrefix
)
```

Arguments:

`resultsConnectionDetails` The connection details to the results database which is an object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`resultsConnectionDetails` The connection details to the results database which is an object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`resultsDatabaseSchema` The schema in the results database that holds the results data model.

`tablePrefix` A prefix to apply to the database table names (optional).

`tablePrefix` A prefix to apply to the database table names (optional).

Method `getResultsDataModelSpecification()`: Get the results data model specification for the module

Usage:

```
CharacterizationModule$getResultsDataModelSpecification(tablePrefix = "")
```

Arguments:

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method uploadResults(): Upload the results for the module

Usage:

```
CharacterizationModule$uploadResults(
  resultsConnectionDetails,
  analysisSpecifications,
  resultsDataModelSettings
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)

resultsDataModelSettings The results data model settings as created using [@seealso createResultsDataModel](#)

Method createModuleSpecifications(): Creates the CharacterizationModule Specifications

Usage:

```
CharacterizationModule$createModuleSpecifications(
  targetIds,
  outcomeIds,
  outcomeWashoutDays = c(365),
  minPriorObservation = 365,
  dechallengeStopInterval = 30,
  dechallengeEvaluationWindow = 30,
  riskWindowStart = c(1, 1),
  startAnchor = c("cohort start", "cohort start"),
  riskWindowEnd = c(0, 365),
  endAnchor = c("cohort end", "cohort end"),
  minCharacterizationMean = 0.01,
  covariateSettings = FeatureExtraction::createCovariateSettings(useDemographicsGender =
    T, useDemographicsAge = T, useDemographicsAgeGroup = T, useDemographicsRace = T,
    useDemographicsEthnicity = T, useDemographicsIndexYear = T, useDemographicsIndexMonth
    = T, useDemographicsTimeInCohort = T, useDemographicsPriorObservationTime = T,
    useDemographicsPostObservationTime = T, useConditionGroupEraLongTerm = T,
    useDrugGroupEraOverlapping = T, useDrugGroupEraLongTerm = T,
    useProcedureOccurrenceLongTerm = T, useMeasurementLongTerm = T,

    useObservationLongTerm = T, useDeviceExposureLongTerm = T,
    useVisitConceptCountLongTerm = T, useConditionGroupEraShortTerm = T,
    useDrugGroupEraShortTerm = T, useProcedureOccurrenceShortTerm = T,
    useMeasurementShortTerm = T, useObservationShortTerm = T, useDeviceExposureShortTerm
```

```

    = T, useVisitConceptCountShortTerm = T, endDays = 0, longTermStartDays = -365,
      shortTermStartDays = -30),
  caseCovariateSettings =
    Characterization::createDuringCovariateSettings(useConditionGroupEraDuring = T,
    useDrugGroupEraDuring = T, useProcedureOccurrenceDuring = T, useDeviceExposureDuring
    = T, useMeasurementDuring = T, useObservationDuring = T, useVisitConceptCountDuring =
      T),
  casePreTargetDuration = 365,
  casePostOutcomeDuration = 365,
  includeTimeToEvent = TRUE,
  includeDechallengeRechallenge = TRUE,
  includeAggregateCovariate = TRUE
)

```

Arguments:

targetIds A vector of cohort IDs to use as the target(s) for the characterization

outcomeIds A vector of cohort IDs to use as the outcome(s) for the characterization

outcomeWashoutDays A vector of integers specifying the washout days for each outcome (same length as the outcomeIds)

minPriorObservation The number of days of minimum observation a patient in the target populations must have

dechallengeStopInterval description

dechallengeEvaluationWindow description

riskWindowStart The number of days after start anchor to start the time-at-risk (can be a vector for multiple TARS)

startAnchor The TAR starts relative to this either cohort start or cohort end (can be a vector for multiple TARS)

riskWindowEnd The number of days after end anchor to end the time-at-risk (can be a vector for multiple TARS)

endAnchor The TAR ends relative to this either cohort start or cohort end (can be a vector for multiple TARS)

minCharacterizationMean The minimum fraction patients in the target have a covariate for it to be included

covariateSettings Covariates for the database, cohort and risk factor characterization

caseCovariateSettings Covariates for the case-series characterization

casePreTargetDuration The number of days before target start to use for case-series

casePostOutcomeDuration The number of days after outcome start to use for case-series

includeTimeToEvent Lets you skip running a time to event analyses when set to FALSE

includeDechallengeRechallenge Lets you skip running a dechallenge-rechallenge analyses when set to FALSE

includeAggregateCovariate Lets you skip running the aggregate covariate analyses when set to FALSE

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

```
CharacterizationModule$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

CohortDiagnosticsModule

Evaluate phenotypes with the R
[hrefhttps://ohdsi.github.io/CohortDiagnostics/HADES](https://ohdsi.github.io/CohortDiagnostics/HADES)
CohortDiagnostics Package

Description

Development and evaluation of phenotype algorithms against the OMOP Common Data Model.

Super class

`Strategus::StrategusModule` -> CohortDiagnosticsModule

Public fields

`tablePrefix` The table prefix to append to results tables

Methods

Public methods:

- `CohortDiagnosticsModule$new()`
- `CohortDiagnosticsModule$execute()`
- `CohortDiagnosticsModule$createResultsDataModel()`
- `CohortDiagnosticsModule$getResultsDataModelSpecification()`
- `CohortDiagnosticsModule$uploadResults()`
- `CohortDiagnosticsModule$createModuleSpecifications()`
- `CohortDiagnosticsModule$validateModuleSpecifications()`
- `CohortDiagnosticsModule$clone()`

Method `new()`: Initialize the module

Usage:

`CohortDiagnosticsModule$new()`

Method `execute()`: Executes the CohortDiagnostics package

Usage:

```
CohortDiagnosticsModule$execute(
  connectionDetails,
  analysisSpecifications,
  executionSettings
)
```

Arguments:

`connectionDetails` An object of class `connectionDetails` as created by the `DatabaseConnector::createConnectionDetails()` function.

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by `createEmptyAnalysisSpecifications()`

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by `createEmptyAnalysisSpecifications()`

`executionSettings` An object of type `ExecutionSettings` as created by `createCdmExecutionSettings()` or `createResultsExecutionSettings()`.

Method createResultsDataModel(): Create the results data model for the module

Usage:

```
CohortDiagnosticsModule$createResultsDataModel(
  resultsConnectionDetails,
  resultsDatabaseSchema,
  tablePrefix = self$tablePrefix
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsDatabaseSchema The schema in the results database that holds the results data model.

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method getResultsDataModelSpecification(): Get the results data model specification for the module

Usage:

```
CohortDiagnosticsModule$getResultsDataModelSpecification(tablePrefix = "")
```

Arguments:

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method uploadResults(): Upload the results for the module

Usage:

```
CohortDiagnosticsModule$uploadResults(
  resultsConnectionDetails,
  analysisSpecifications,
  resultsDataModelSettings
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)

resultsDataModelSettings The results data model settings as created using [[@seealso createResultsDataModel](#)]

Method createModuleSpecifications(): Creates the CohortDiagnostics Module Specifications

Usage:

```

CohortDiagnosticsModule$createModuleSpecifications(
  cohortIds = NULL,
  runInclusionStatistics = TRUE,
  runIncludedSourceConcepts = TRUE,
  runOrphanConcepts = TRUE,
  runTimeSeries = FALSE,
  runVisitContext = TRUE,
  runBreakdownIndexEvents = TRUE,
  runIncidenceRate = TRUE,
  runCohortRelationship = TRUE,
  runTemporalCohortCharacterization = TRUE,
  temporalCovariateSettings = private$.getDefaultCovariateSettings(),
  minCharacterizationMean = 0.01,
  irWashoutPeriod = 0
)

```

Arguments:

cohortIds A list of cohort IDs to use when running the CohortDiagnostics. Default is NULL which will use all cohorts present in the cohort definition set in the analysis specification

runInclusionStatistics Generate and export statistic on the cohort inclusion rules?

runIncludedSourceConcepts Generate and export the source concepts included in the cohorts?

runOrphanConcepts Generate and export potential orphan concepts?

runTimeSeries Generate and export the time series diagnostics?

runVisitContext Generate and export index-date visit context?

runBreakdownIndexEvents Generate and export the breakdown of index events?

runIncidenceRate Generate and export the cohort incidence rates?

runCohortRelationship Generate and export the cohort relationship? Cohort relationship checks the temporal relationship between two or more cohorts.

runTemporalCohortCharacterization Generate and export the temporal cohort characterization? Only records with values greater than 0.001 are returned.

temporalCovariateSettings Either an object of type `covariateSettings` as created using one of the `createTemporalCovariateSettings` function in the `FeatureExtraction` package, or a list of such objects.

minCharacterizationMean The minimum mean value for characterization output. Values below this will be cut off from output. This will help reduce the file size of the characterization output, but will remove information on covariates that have very low values. The default is 0.001 (i.e. 0.1 percent)

irWashoutPeriod Number of days washout to include in calculation of incidence rates - default is 0

Method `validateModuleSpecifications()`: Validate the module specifications

Usage:

```
CohortDiagnosticsModule$validateModuleSpecifications(moduleSpecifications)
```

Arguments:

moduleSpecifications The CohortIncidence module specifications

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

```
CohortDiagnosticsModule$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

CohortGeneratorModule *Generate cohorts with the R* [hrefhttps://ohdsi.github.io/CohortGenerator/HADES](https://ohdsi.github.io/CohortGenerator/HADES)
CohortGenerator Package

Description

Generates cohorts against the OMOP Common Data Model

Super class

`Strategus::StrategusModule` -> CohortGeneratorModule

Public fields

cohortDefinitionSharedResourcesClassName A constant for the name of the cohort definition shared resources section of the analysis specification

negativeControlOutcomeSharedResourcesClassName A constant for the name of the negative control outcome shared resources section of the analysis specification

Methods**Public methods:**

- `CohortGeneratorModule$new()`
- `CohortGeneratorModule$execute()`
- `CohortGeneratorModule$createResultsDataModel()`
- `CohortGeneratorModule$getResultsDataModelSpecification()`
- `CohortGeneratorModule$uploadResults()`
- `CohortGeneratorModule$createModuleSpecifications()`
- `CohortGeneratorModule$createCohortSharedResourceSpecifications()`
- `CohortGeneratorModule$createNegativeControlOutcomeCohortSharedResourceSpecifications()`
- `CohortGeneratorModule$validateModuleSpecifications()`
- `CohortGeneratorModule$validateCohortSharedResourceSpecifications()`
- `CohortGeneratorModule$validateNegativeControlOutcomeCohortSharedResourceSpecifications()`
- `CohortGeneratorModule$clone()`

Method `new()`: Initialize the module

Usage:

```
CohortGeneratorModule$new()
```

Method `execute()`: Generates the cohorts

Usage:

```
CohortGeneratorModule$execute(
  connectionDetails,
  analysisSpecifications,
  executionSettings
)
```


Arguments:

connectionDetails An object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications\(\)](#)

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications\(\)](#)

executionSettings An object of type ExecutionSettings as created by [createCdmExecutionSettings\(\)](#) or [createResultsExecutionSettings\(\)](#).

Method createResultsDataModel(): Create the results data model for the module

Usage:

```
CohortGeneratorModule$createResultsDataModel(
  resultsConnectionDetails,
  resultsDatabaseSchema,
  tablePrefix = ""
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsDatabaseSchema The schema in the results database that holds the results data model.

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method getResultsDataModelSpecification(): Get the results data model specification for the module

Usage:

```
CohortGeneratorModule$getResultsDataModelSpecification(tablePrefix = "")
```

Arguments:

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method uploadResults(): Upload the results for the module

Usage:

```
CohortGeneratorModule$uploadResults(
  resultsConnectionDetails,
  analysisSpecifications,
  resultsDataModelSettings
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)
 analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)
 resultsDataModelSettings The results data model settings as created using [@seealso createResultsDataModel](#)

Method createModuleSpecifications(): Creates the CohortGenerator Module Specifications

Usage:

```
CohortGeneratorModule$createModuleSpecifications(generateStats = TRUE)
```

Arguments:

generateStats When TRUE, the Circe cohort definition SQL will include steps to compute inclusion rule statistics.

Method createCohortSharedResourceSpecifications(): Create shared specifications for the cohort definition set

Usage:

```
CohortGeneratorModule$createCohortSharedResourceSpecifications(  
  cohortDefinitionSet  
)
```

Arguments:

cohortDefinitionSet The cohort definition set to include in the specification. See the CohortGenerator package for details on how to build this object.

Method createNegativeControlOutcomeCohortSharedResourceSpecifications(): Create shared specifications for the negative control outcomes cohort set

Usage:

```
CohortGeneratorModule$createNegativeControlOutcomeCohortSharedResourceSpecifications(  
  negativeControlOutcomeCohortSet,  
  occurrenceType,  
  detectOnDescendants  
)
```

Arguments:

negativeControlOutcomeCohortSet The negative control outcome cohort definition set defines the concepts to use to construct negative control outcome cohorts. See the CohortGenerator package for more details.

occurrenceType Either "first" or "all"

detectOnDescendants When TRUE, the concept ID for the negative control will use the concept_ancestor table and will detect descendant concepts when constructing the cohort.

Method validateModuleSpecifications(): Validate the module specifications

Usage:

```
CohortGeneratorModule$validateModuleSpecifications(moduleSpecifications)
```

Arguments:

moduleSpecifications The CohortGenerator module specifications

Method validateCohortSharedResourceSpecifications(): Validate the cohort shared resource specifications

Usage:

```
CohortGeneratorModule$validateCohortSharedResourceSpecifications(
  cohortSharedResourceSpecifications
)
```

Arguments:

cohortSharedResourceSpecifications The cohort shared resource specifications

Method validateNegativeControlOutcomeCohortSharedResourceSpecifications(): Validate the cohort shared resource specifications

Usage:

```
CohortGeneratorModule$validateNegativeControlOutcomeCohortSharedResourceSpecifications(
  negativeControlOutcomeCohortSharedResourceSpecifications
)
```

Arguments:

negativeControlOutcomeCohortSharedResourceSpecifications The cohort shared resource specifications

Method clone(): The objects of this class are cloneable with this method.

Usage:

```
CohortGeneratorModule$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

CohortIncidenceModule *Compute incidence with the R*
[hrefhttps://ohdsi.github.io/CohortIncidence/HADES](https://ohdsi.github.io/CohortIncidence/HADES)
CohortIncidence Package

Description

Computes incidence rates for cohorts against the OMOP Common Data Model

Super class

`Strategus::StrategusModule` -> CohortIncidenceModule

Public fields

tablePrefix The table prefix to append to results tables

Methods

Public methods:

- `CohortIncidenceModule$new()`
- `CohortIncidenceModule$execute()`
- `CohortIncidenceModule$createResultsDataModel()`
- `CohortIncidenceModule$getResultsDataModelSpecification()`
- `CohortIncidenceModule$uploadResults()`
- `CohortIncidenceModule$createModuleSpecifications()`
- `CohortIncidenceModule$validateModuleSpecifications()`

- [CohortIncidenceModule\\$clone\(\)](#)

Method `new()`: Initialize the module

Usage:

```
CohortIncidenceModule$new()
```

Method `execute()`: Execute the CohortIncidence package

Usage:

```
CohortIncidenceModule$execute(
  connectionDetails,
  analysisSpecifications,
  executionSettings
)
```

Arguments:

`connectionDetails` An object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpecifications\(\)](#)

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpecifications\(\)](#)

`executionSettings` An object of type `ExecutionSettings` as created by [createCdmExecutionSettings\(\)](#) or [createResultsExecutionSettings\(\)](#).

Method `createResultsDataModel()`: Create the results data model for the module

Usage:

```
CohortIncidenceModule$createResultsDataModel(
  resultsConnectionDetails,
  resultsDatabaseSchema,
  tablePrefix = ""
)
```

Arguments:

`resultsConnectionDetails` The connection details to the results database which is an object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`resultsConnectionDetails` The connection details to the results database which is an object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`resultsDatabaseSchema` The schema in the results database that holds the results data model.

`tablePrefix` A prefix to apply to the database table names (optional).

`tablePrefix` A prefix to apply to the database table names (optional).

Method `getResultsDataModelSpecification()`: Get the results data model specification for the module

Usage:

```
CohortIncidenceModule$getResultsDataModelSpecification(tablePrefix = "")
```

Arguments:

`tablePrefix` A prefix to apply to the database table names (optional).

`tablePrefix` A prefix to apply to the database table names (optional).

Method `uploadResults()`: Upload the results for the module

Usage:

```
CohortIncidenceModule$uploadResults(
  resultsConnectionDetails,
  analysisSpecifications,
  resultsDataModelSettings
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class `connectionDetails` as created by the `DatabaseConnector::createConnectionDetails()` function.

resultsConnectionDetails The connection details to the results database which is an object of class `connectionDetails` as created by the `DatabaseConnector::createConnectionDetails()` function.

analysisSpecifications An object of type `AnalysisSpecifications` as created by `createEmptyAnalysisSpec`

analysisSpecifications An object of type `AnalysisSpecifications` as created by `createEmptyAnalysisSpec`

resultsDataModelSettings The results data model settings as created using [@seealso `createResultsDataModel`

Method `createModuleSpecifications()`: Creates the CohortIncidence Module Specifications

Usage:

```
CohortIncidenceModule$createModuleSpecifications(irDesign = NULL)
```

Arguments:

irDesign The incidence rate design created from the CohortIncidence package

Method `validateModuleSpecifications()`: Validate the module specifications

Usage:

```
CohortIncidenceModule$validateModuleSpecifications(moduleSpecifications)
```

Arguments:

moduleSpecifications The CohortIncidence module specifications

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

```
CohortIncidenceModule$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

CohortMethodModule	<i>New-user</i>	<i>cohort</i>	<i>studies</i>	<i>with</i>	<i>the</i>
	Rhrefhttps://ohdsi.github.io/CohortMethod/HADES				<i>CohortMethod</i>
	<i>Package</i>				

Description

Module for performing new-user cohort studies against the OMOP Common Data Model

Super class

`Strategus::StrategusModule` -> CohortMethodModule

Methods

Public methods:

- [CohortMethodModule\\$new\(\)](#)
- [CohortMethodModule\\$execute\(\)](#)
- [CohortMethodModule\\$createResultsDataModel\(\)](#)
- [CohortMethodModule\\$getResultsDataModelSpecification\(\)](#)
- [CohortMethodModule\\$uploadResults\(\)](#)
- [CohortMethodModule\\$createModuleSpecifications\(\)](#)
- [CohortMethodModule\\$validateModuleSpecifications\(\)](#)
- [CohortMethodModule\\$clone\(\)](#)

Method `new()`: Initialize the module

Usage:

```
CohortMethodModule$new()
```

Method `execute()`: Executes the CohortMethod package

Usage:

```
CohortMethodModule$execute(
  connectionDetails,
  analysisSpecifications,
  executionSettings
)
```

Arguments:

`connectionDetails` An object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`analysisSpecifications` The analysis specifications for the study

`executionSettings` An object of type `ExecutionSettings` as created by [createCdmExecutionSettings\(\)](#) or [createResultsExecutionSettings\(\)](#).

Method `createResultsDataModel()`: Create the results data model for the module

Usage:

```
CohortMethodModule$createResultsDataModel(
  resultsConnectionDetails,
  resultsDatabaseSchema,
  tablePrefix = ""
)
```

Arguments:

`resultsConnectionDetails` The connection details to the results database which is an object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`resultsDatabaseSchema` The schema in the results database that holds the results data model.

`tablePrefix` A prefix to apply to the database table names (optional).

`tablePrefix` A prefix to apply to the database table names (optional).

Method `getResultsDataModelSpecification()`: Get the results data model specification for the module

Usage:

```
CohortMethodModule$getResultsDataModelSpecification(tablePrefix = "")
```

Arguments:

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method uploadResults(): Upload the results for the module

Usage:

```
CohortMethodModule$uploadResults(
  resultsConnectionDetails,
  analysisSpecifications,
  resultsDataModelSettings
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)

resultsDataModelSettings The results data model settings as created using [[@seealso createResultsDataModel](#)]

Method createModuleSpecifications(): Creates the CohortMethod Module Specifications

Usage:

```
CohortMethodModule$createModuleSpecifications(
  cmAnalysisList,
  targetComparatorOutcomesList,
  analysesToExclude = NULL,
  refitPsForEveryOutcome = FALSE,
  refitPsForEveryStudyPopulation = TRUE,
  cmDiagnosticThresholds = CohortMethod::createCmDiagnosticThresholds()
)
```

Arguments:

cmAnalysisList A list of objects of type cmAnalysis as created using the '[CohortMethod::createCmAnalysis](#)' function.

targetComparatorOutcomesList A list of objects of type targetComparatorOutcomes as created using the [CohortMethod::createTargetComparatorOutcomes](#) function.

analysesToExclude Analyses to exclude. See the Analyses to Exclude section for details.

refitPsForEveryOutcome Should the propensity model be fitted for every outcome (i.e. after people who already had the outcome are removed)? If false, a single propensity model will be fitted, and people who had the outcome previously will be removed afterwards.

refitPsForEveryStudyPopulation Should the propensity model be fitted for every study population definition? If false, a single propensity model will be fitted, and the study population criteria will be applied afterwards.

cmDiagnosticThresholds An object of type CmDiagnosticThresholds as created using [CohortMethod::createCmDiagnosticThresholds](#)

Details: Run a list of analyses for the target-comparator-outcomes of interest. This function will run all specified analyses against all hypotheses of interest, meaning that the total number of

outcome models is `length(cmAnalysisList) * length(targetComparatorOutcomesList)` (if all analyses specify an outcome model should be fitted). When you provide several analyses it will determine whether any of the analyses have anything in common, and will take advantage of this fact. For example, if we specify several analyses that only differ in the way the outcome model is fitted, then this function will extract the data and fit the propensity model only once, and re-use this in all the analysis.

After completion, a tibble containing references to all generated files can be obtained using the `CohortMethod::getFileReference()` function. A summary of the analysis results can be obtained using the `CohortMethod::getResultsSummary()` function.

Analyses to Exclude:

Normally, `runCmAnalyses` will run all combinations of target-comparator-outcome-analyses settings. However, sometimes we may not need all those combinations. Using the `analysesToExclude` argument, we can remove certain items from the full matrix. This argument should be a data frame with at least one of the following columns:

Method `validateModuleSpecifications()`: Validate the module specifications

Usage:

```
CohortMethodModule$validateModuleSpecifications(moduleSpecifications)
```

Arguments:

`moduleSpecifications` The CohortMethod module specifications

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

```
CohortMethodModule$clone(deep = FALSE)
```

Arguments:

`deep` Whether to make a deep clone.

`createCdmExecutionSettings`

Create CDM execution settings

Description

Create CDM execution settings

Usage

```
createCdmExecutionSettings(
  workDatabaseSchema,
  cdmDatabaseSchema,
  cohortTableNames = CohortGenerator::getCohortTableNames(cohortTable = "cohort"),
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  workFolder,
  resultsFolder,
  logFileName = file.path(resultsFolder, "strategus-log.txt"),
  minCellCount = 5,
  incremental = TRUE,
  maxCores = parallel::detectCores(),
  modulesToExecute = c()
)
```


Arguments**workDatabaseSchema**

A database schema where intermediate data can be stored. The user (as identified in the connection details) will need to have write access to this database schema.

cdmDatabaseSchema

The database schema containing the data in CDM format. The user (as identified in the connection details) will need to have read access to this database schema.

cohortTableNames

An object identifying the various cohort table names that will be created in the workDatabaseSchema. This object can be created using the [CohortGenerator::getCohortTableNames](#) function.

tempEmulationSchema

Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

workFolder

A folder in the local file system where intermediate results can be written.

resultsFolder

The root folder holding the study results.

logFileName

Logging information from Strategus and all modules will be located in this file. Individual modules will continue to have their own module-specific logs. By default this will be written to the root of the resultsFolder

minCellCount

The minimum number of subjects contributing to a count before it can be included in results.

incremental

This value will be passed to each module that supports execution in an incremental manner. Modules and their underlying packages may use the workFolder contents to determine their state of execution and attempt to pick up where they left off when this value is set to TRUE.

maxCores

The maximum number of processing cores to use for execution. The default is to use all available cores on the machine.

modulesToExecute

(Optional) A vector with the list of modules to execute. When an empty vector/NULL is supplied (default), all modules in the analysis specification are executed.

Value

An object of type ExecutionSettings.

```
createEmptyAnalysisSpecifications
```

Create an empty analysis specifications object.

Description

Create an empty analysis specifications object.

Usage

```
createEmptyAnalysisSpecifications()
```

Value

An object of type AnalysisSpecifications.

```
createEmptyAnalysisSpecifications
```

DEPRECATED DUE TO MISSPELLING

Description

DEPRECATED DUE TO MISSPELLING

Usage

```
createEmptyAnalysisSpecifications()
```

See Also

[createEmptyAnalysisSpecifications\(\)](#)

```
createResultDataModel
```

Create Result Data Model

Description

Use this at the study design stage to create data models for modules. This function loads modules and executes any custom code to create the results data model in the specified schema in the results database.

Usage

```
createResultDataModel(
  analysisSpecifications,
  resultsDataModelSettings,
  resultsConnectionDetails
)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications\(\)](#)

resultsDataModelSettings

The results data model settings as created using [[@seealso createResultsDataModelSettings\(\)](#)]

resultsConnectionDetails

The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

```
createResultsDataModelSettings
```

Create Results Data Model Settings

Description

The results data model settings are used to create the results data model and to upload results.

Usage

```
createResultsDataModelSettings(
  resultsDatabaseSchema,
  resultsFolder,
  logFileName = file.path(resultsFolder, "strategus-results-data-model-log.txt"),
  modulesToExecute = c()
)
```

Arguments

resultsDatabaseSchema

The schema in the results database that holds the results data model.

resultsFolder The root folder holding the study results.

logFileName Log location for data model operations

modulesToExecute

(Optional) A vector with the list of modules to execute. When an empty vector/NULL is supplied (default), all modules in the analysis specification are executed.

Value

An object of type ResultsDataModelSettings

```
createResultsExecutionSettings
```

Create Results execution settings

Description

Create Results execution settings

Usage

```
createResultsExecutionSettings(
  resultsDatabaseSchema,
  workFolder,
  resultsFolder,
  logFileName = file.path(resultsFolder, "strategus-log.txt"),
  minCellCount = 5,
  maxCores = parallel::detectCores(),
  modulesToExecute = c()
)
```

Arguments

resultsDatabaseSchema	The schema in the results database that holds the results data model.
workFolder	A folder in the local file system where intermediate results can be written.
resultsFolder	The root folder holding the study results.
logFileName	Logging information from Strategus and all modules will be located in this file. Individual modules will continue to have their own module-specific logs. By default this will be written to the root of the resultsFolder
minCellCount	The minimum number of subjects contributing to a count before it can be included in results.
maxCores	The maximum number of processing cores to use for execution. The default is to use all available cores on the machine.
modulesToExecute	(Optional) A vector with the list of modules to execute. When an empty vector/NULL is supplied (default), all modules in the analysis specification are executed.

Value

An object of type ExecutionSettings.

EvidenceSynthesisModule

Meta-analysis with the R [hrefhttps://ohdsi.github.io/EvidenceSynthesis/HADES](https://ohdsi.github.io/EvidenceSynthesis/HADES)
EvidenceSynthesis Package

Description

Module for for combining causal effect estimates and study diagnostics across multiple data sites in a distributed study. This includes functions for performing meta-analysis and forest plots

Super class

`Strategus::StrategusModule` -> EvidenceSynthesisModule

Methods**Public methods:**

- `EvidenceSynthesisModule$new()`
- `EvidenceSynthesisModule$execute()`
- `EvidenceSynthesisModule$createResultsDataModel()`
- `EvidenceSynthesisModule$getResultsDataModelSpecification()`
- `EvidenceSynthesisModule$uploadResults()`
- `EvidenceSynthesisModule$validateModuleSpecifications()`
- `EvidenceSynthesisModule$createEvidenceSynthesisSource()`
- `EvidenceSynthesisModule$createRandomEffectsMetaAnalysis()`
- `EvidenceSynthesisModule$createFixedEffectsMetaAnalysis()`
- `EvidenceSynthesisModule$createBayesianMetaAnalysis()`

- [EvidenceSynthesisModule\\$createEsDiagnosticThresholds\(\)](#)
- [EvidenceSynthesisModule\\$createModuleSpecifications\(\)](#)
- [EvidenceSynthesisModule\\$clone\(\)](#)

Method `new()`: Initialize the module

Usage:

```
EvidenceSynthesisModule$new()
```

Method `execute()`: Executes the EvidenceSynthesis package

Usage:

```
EvidenceSynthesisModule$execute(
  connectionDetails,
  analysisSpecifications,
  executionSettings
)
```

Arguments:

`connectionDetails` An object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpecifications\(\)](#)

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpecifications\(\)](#)

`executionSettings` An object of type `ExecutionSettings` as created by [createCdmExecutionSettings\(\)](#) or [createResultsExecutionSettings\(\)](#).

Method `createResultsDataModel()`: Create the results data model for the module

Usage:

```
EvidenceSynthesisModule$createResultsDataModel(
  resultsConnectionDetails,
  resultsDatabaseSchema,
  tablePrefix = ""
)
```

Arguments:

`resultsConnectionDetails` The connection details to the results database which is an object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`resultsConnectionDetails` The connection details to the results database which is an object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`resultsDatabaseSchema` The schema in the results database that holds the results data model.

`tablePrefix` A prefix to apply to the database table names (optional).

`tablePrefix` A prefix to apply to the database table names (optional).

Method `getResultsDataModelSpecification()`: Get the results data model specification for the module

Usage:

```
EvidenceSynthesisModule$getResultsDataModelSpecification(tablePrefix = "")
```

Arguments:

`tablePrefix` A prefix to apply to the database table names (optional).

`tablePrefix` A prefix to apply to the database table names (optional).

Method uploadResults(): Upload the results for the module

Usage:

```
EvidenceSynthesisModule$uploadResults(
  resultsConnectionDetails,
  analysisSpecifications,
  resultsDataModelSettings
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)

resultsDataModelSettings The results data model settings as created using [[@seealso createResultsDataModel](#)]

Method validateModuleSpecifications(): Validate the module specifications

Usage:

```
EvidenceSynthesisModule$validateModuleSpecifications(moduleSpecifications)
```

Arguments:

moduleSpecifications The EvidenceSynthesis module specifications Create an evidence synthesis source

Method createEvidenceSynthesisSource():

Usage:

```
EvidenceSynthesisModule$createEvidenceSynthesisSource(
  sourceMethod = "CohortMethod",
  databaseIds = NULL,
  analysisIds = NULL,
  likelihoodApproximation = "grid with gradients"
)
```

Arguments:

sourceMethod The source method generating the estimates to synthesize. Can be "Cohort-Method" or "SelfControlledCaseSeries"

databaseIds The database IDs to include. Use databaseIds = NULL to include all database IDs.

analysisIds The source method analysis IDs to include. Use analysisIds = NULL to include all analysis IDs.

likelihoodApproximation The type of likelihood approximation. Can be "adaptive grid", "normal", or "grid with gradients".

Returns: An object of type EvidenceSynthesisSource. Create parameters for a random-effects meta-analysis

Method createRandomEffectsMetaAnalysis():

Usage:

```
EvidenceSynthesisModule$createRandomEffectsMetaAnalysis(
  alpha = 0.05,
  evidenceSynthesisAnalysisId = 1,
  evidenceSynthesisDescription = "Random-effects",
  evidenceSynthesisSource = NULL,
  controlType = "outcome"
)
```

Arguments:

alpha The alpha (expected type I error) used for the confidence intervals.
 evidenceSynthesisAnalysisId description
 evidenceSynthesisDescription description
 evidenceSynthesisSource description
 controlType description Create a parameter object for the function computeFixedEffectMeta-Analysis

Details: Use DerSimonian-Laird meta-analysis

Method createFixedEffectsMetaAnalysis():

Usage:

```
EvidenceSynthesisModule$createFixedEffectsMetaAnalysis(
  alpha = 0.05,
  evidenceSynthesisAnalysisId = 1,
  evidenceSynthesisDescription = "Fixed-effects",
  evidenceSynthesisSource = NULL,
  controlType = "outcome"
)
```

Arguments:

alpha The alpha (expected type I error) used for the confidence intervals.
 evidenceSynthesisAnalysisId description
 evidenceSynthesisDescription description
 evidenceSynthesisSource description
 controlType description Create a parameter object for the function computeBayesianMeta-Analysis

Details: Create an object defining the parameter values.

Method createBayesianMetaAnalysis():

Usage:

```
EvidenceSynthesisModule$createBayesianMetaAnalysis(
  chainLength = 1100000,
  burnIn = 1e+05,
  subSampleFrequency = 100,
  priorSd = c(2, 0.5),
  alpha = 0.05,
  robust = FALSE,
  df = 4,
  seed = 1,
  evidenceSynthesisAnalysisId = 1,
  evidenceSynthesisDescription = "Bayesian random-effects",
  evidenceSynthesisSource = NULL,
  controlType = "outcome"
)
```

Arguments:

chainLength Number of MCMC iterations.
 burnIn Number of MCMC iterations to consider as burn in.
 subSampleFrequency Subsample frequency for the MCMC.
 priorSd A two-dimensional vector with the standard deviation of the prior for mu and tau, respectively.
 alpha The alpha (expected type I error) used for the credible intervals.
 robust Whether or not to use a t-distribution model; default: FALSE.
 df Degrees of freedom for the t-model, only used if robust is TRUE.
 seed The seed for the random number generator.
 evidenceSynthesisAnalysisId description
 evidenceSynthesisDescription description
 evidenceSynthesisSource description
 controlType description Create EvidenceSynthesis diagnostics thresholds

Details: Create an object defining the parameter values.

Method createEsDiagnosticThresholds(): Threshold used to determine if we pass or fail diagnostics.

Usage:

```
EvidenceSynthesisModule$createEsDiagnosticThresholds(
  mdrThreshold = 10,
  easeThreshold = 0.25,
  i2Threshold = 0.4,
  tauThreshold = log(2)
)
```

Arguments:

mdrThreshold What is the maximum allowed minimum detectable relative risk (MDRR)?
 easeThreshold What is the maximum allowed expected absolute systematic error (EASE).
 i2Threshold What is the maximum allowed I^2 (measure of between-database heterogeneity in random-effects models)?
 tauThreshold What is the maximum allowed tau (measure of between-database heterogeneity in Bayesian random-effects models)?

Returns: An object of type EsDiagnosticThresholds.

Method createModuleSpecifications(): Creates the module Specifications

Usage:

```
EvidenceSynthesisModule$createModuleSpecifications(
  evidenceSynthesisAnalysisList,
  esDiagnosticThresholds = self$createEsDiagnosticThresholds()
)
```

Arguments:

evidenceSynthesisAnalysisList A list of objects of type EvidenceSynthesisAnalysis as generated by either the `EvidenceSynthesisModule$createFixedEffectsMetaAnalysis()` or `EvidenceSynthesisModule$createBayesianMetaAnalysis()` function.
 esDiagnosticThresholds An object of type EsDiagnosticThresholds as generated by the `EvidenceSynthesisModule$createEsDiagnosticThresholds()` function.

Method clone(): The objects of this class are cloneable with this method.

Usage:

```
EvidenceSynthesisModule$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

 execute

Execute analysis specifications.

Description

Execute analysis specifications.

Usage

```
execute(analysisSpecifications, executionSettings, connectionDetails)
```

Arguments

analysisSpecifications

An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications\(\)](#)

executionSettings

An object of type ExecutionSettings as created by [createCdmExecutionSettings\(\)](#) or [createResultsExecutionSettings\(\)](#).

connectionDetails

An object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

Value

Returns a list of lists that contains

- moduleName: The name of the module executed
- result: The result of the execution. See [purrr::safely](#) for details on this result.
- executionTime: The time for the module to execute

 getCdmDatabaseMetaData

Gets the metadata for your OMOP CDM Database

Description

This function is used to gather metadata about your OMOP CDM and inspect for informational purposes. This information will be saved with your results when executing an analysis specification.

Usage

```
getCdmDatabaseMetaData(cdmExecutionSettings, connectionDetails)
```

Arguments

cdmExecutionSettings

An object of type CdmExecutionSettings as created [createCdmExecutionSettings\(\)](#).

connectionDetails

An object of class connectionDetails as created by the [DatabaseConnector::createConnection](#) function.

PatientLevelPredictionModule

Patient-level prediction with the R
[hrefhttps://ohdsi.github.io/PatientLevelPrediction/HADES](https://ohdsi.github.io/PatientLevelPrediction/HADES)
PatientLevelPrediction Package

Description

Module for performing patient-level prediction in an observational database in the OMOP Common Data Model.

Super class

[Strategus::StrategusModule](#) -> PatientLevelPredictionModule

Public fields

tablePrefix The table prefix to append to the results tables

Methods**Public methods:**

- [PatientLevelPredictionModule\\$new\(\)](#)
- [PatientLevelPredictionModule\\$execute\(\)](#)
- [PatientLevelPredictionModule\\$createResultsDataModel\(\)](#)
- [PatientLevelPredictionModule\\$getResultsDataModelSpecification\(\)](#)
- [PatientLevelPredictionModule\\$uploadResults\(\)](#)
- [PatientLevelPredictionModule\\$createModuleSpecifications\(\)](#)
- [PatientLevelPredictionModule\\$validateModuleSpecifications\(\)](#)
- [PatientLevelPredictionModule\\$clone\(\)](#)

Method new(): Initialize the module

Usage:

```
PatientLevelPredictionModule$new()
```

Method execute(): Executes the PatientLevelPrediction package

Usage:

```
PatientLevelPredictionModule$execute(
  connectionDetails,
  analysisSpecifications,
  executionSettings
)
```

Arguments:

connectionDetails An object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications\(\)](#)

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications\(\)](#)

executionSettings An object of type ExecutionSettings as created by [createCdmExecutionSettings\(\)](#) or [createResultsExecutionSettings\(\)](#).

Method createResultsDataModel(): Create the results data model for the module

Usage:

```
PatientLevelPredictionModule$createResultsDataModel(
  resultsConnectionDetails,
  resultsDatabaseSchema,
  tablePrefix = self$tablePrefix
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsDatabaseSchema The schema in the results database that holds the results data model.

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method getResultsDataModelSpecification(): Get the results data model specification for the module

Usage:

```
PatientLevelPredictionModule$getResultsDataModelSpecification(tablePrefix = "")
```

Arguments:

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method uploadResults(): Upload the results for the module

Usage:

```
PatientLevelPredictionModule$uploadResults(
  resultsConnectionDetails,
  analysisSpecifications,
  resultsDataModelSettings
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)
 analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)
 resultsDataModelSettings The results data model settings as created using [@seealso [createResultsDataModel](#)

Method createModuleSpecifications(): Creates the PatientLevelPrediction Module Specifications

Usage:

```
PatientLevelPredictionModule$createModuleSpecifications(  
  modelDesignList,  
  skipDiagnostics = FALSE  
)
```

Arguments:

modelDesignList A list of model designs created using PatientLevelPrediction::createModelDesign()
 skipDiagnostics Whether to run the diagnostics based on PROBAST before model development

Method validateModuleSpecifications(): Validate the module specifications

Usage:

```
PatientLevelPredictionModule$validateModuleSpecifications(moduleSpecifications)
```

Arguments:

moduleSpecifications The PatientLevelPrediction module specifications

Method clone(): The objects of this class are cloneable with this method.

Usage:

```
PatientLevelPredictionModule$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

PatientLevelPredictionValidationModule

*Validation of patient-level prediction models with the
 R[hrefhttps://ohdsi.github.io/PatientLevelPrediction/HADES](https://ohdsi.github.io/PatientLevelPrediction/HADES) Pa-
 tientLevelPrediction Package*

Description

Module for performing patient-level prediction model validation for models built using the PatientLevelPrediction package.

Super class

[Strategus::StrategusModule](#) -> PatientLevelPredictionValidationModule

Public fields

tablePrefix The table prefix to append to the results tables

Methods

Public methods:

- [PatientLevelPredictionValidationModule\\$new\(\)](#)
- [PatientLevelPredictionValidationModule\\$execute\(\)](#)
- [PatientLevelPredictionValidationModule\\$createResultsDataModel\(\)](#)
- [PatientLevelPredictionValidationModule\\$uploadResults\(\)](#)
- [PatientLevelPredictionValidationModule\\$createModuleSpecifications\(\)](#)
- [PatientLevelPredictionValidationModule\\$validateModuleSpecifications\(\)](#)
- [PatientLevelPredictionValidationModule\\$clone\(\)](#)

Method `new()`: Initialize the module

Usage:

```
PatientLevelPredictionValidationModule$new()
```

Method `execute()`: Executes the PatientLevelPrediction package to validate a PLP model

Usage:

```
PatientLevelPredictionValidationModule$execute(
  connectionDetails,
  analysisSpecifications,
  executionSettings
)
```

Arguments:

`connectionDetails` An object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpecifications\(\)](#)

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpecifications\(\)](#)

`executionSettings` An object of type `ExecutionSettings` as created by [createCdmExecutionSettings\(\)](#) or [createResultsExecutionSettings\(\)](#).

Method `createResultsDataModel()`: Create the results data model for the module

Usage:

```
PatientLevelPredictionValidationModule$createResultsDataModel(
  resultsConnectionDetails,
  resultsDatabaseSchema,
  tablePrefix = self$tablePrefix
)
```

Arguments:

`resultsConnectionDetails` The connection details to the results database which is an object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`resultsConnectionDetails` The connection details to the results database which is an object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`resultsDatabaseSchema` The schema in the results database that holds the results data model.

`tablePrefix` A prefix to apply to the database table names (optional).

Method `uploadResults()`: Upload the results for the module

Usage:

```
PatientLevelPredictionValidationModule$uploadResults(
  resultsConnectionDetails,
  analysisSpecifications,
  resultsDataModelSettings
)
```

Arguments:

`resultsConnectionDetails` The connection details to the results database which is an object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`resultsConnectionDetails` The connection details to the results database which is an object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpec](#)

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpec](#)

`resultsDataModelSettings` The results data model settings as created using [[@seealso createResultsDataModel](#)]

Method `createModuleSpecifications()`: Creates the PatientLevelPredictionValidation Module Specifications

Usage:

```
PatientLevelPredictionValidationModule$createModuleSpecifications(
  validationList = list(PatientLevelPrediction::createValidationDesign(plpModelList =
    list(file.path("location_to_model")), targetId = 1, outcomeId = 3,
    restrictPlpDataSettings = PatientLevelPrediction::createRestrictPlpDataSettings(),
    populationSettings = NULL, recalibrate = "weakRecalibration", runCovariateSummary =
    TRUE), PatientLevelPrediction::createValidationDesign(plpModelList =
    list(file.path("location_to_model")), targetId = 4, outcomeId = 3,
    restrictPlpDataSettings = PatientLevelPrediction::createRestrictPlpDataSettings(),

    populationSettings = NULL, recalibrate = "weakRecalibration", runCovariateSummary
    = TRUE)),
  logLevel = "INFO"
)
```

Arguments:

`validationList` A list of validation designs from `PatientLevelPrediction::createValidationDesign`

`logLevel` The logging level while executing the model validation.

Method `validateModuleSpecifications()`: Validate the module specifications

Usage:

```
PatientLevelPredictionValidationModule$validateModuleSpecifications(
  moduleSpecifications
)
```

Arguments:

`moduleSpecifications` The PatientLevelPredictionValidation module specifications

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

```
PatientLevelPredictionValidationModule$clone(deep = FALSE)
```

Arguments:

`deep` Whether to make a deep clone.

SelfControlledCaseSeriesModule

*Self-Controlled Case Series design with the
[Rhrefhttps://ohdsi.github.io/SelfControlledCaseSeries/HADES](https://ohdsi.github.io/SelfControlledCaseSeries/HADES)
SelfControlledCaseSeries Package*

Description

Module for performing Self-Controlled Case Series (SCCS) analyses against the OMOP Common Data Model.

Super class

`Strategus::StrategusModule` -> SelfControlledCaseSeriesModule

Public fields

`tablePrefix` The table prefix for results tables

Methods

Public methods:

- `SelfControlledCaseSeriesModule$new()`
- `SelfControlledCaseSeriesModule$execute()`
- `SelfControlledCaseSeriesModule$createResultsDataModel()`
- `SelfControlledCaseSeriesModule$getResultsDataModelSpecification()`
- `SelfControlledCaseSeriesModule$uploadResults()`
- `SelfControlledCaseSeriesModule$createModuleSpecifications()`
- `SelfControlledCaseSeriesModule$validateModuleSpecifications()`
- `SelfControlledCaseSeriesModule$clone()`

Method `new()`: Initialize the module

Usage:

```
SelfControlledCaseSeriesModule$new()
```

Method `execute()`: Executes the SelfControlledCaseSeries package

Usage:

```
SelfControlledCaseSeriesModule$execute(  
  connectionDetails,  
  analysisSpecifications,  
  executionSettings  
)
```

Arguments:

`connectionDetails` An object of class `connectionDetails` as created by the `DatabaseConnector::createConnectionDetails` function.

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by `createEmptyAnalysisSpecifications`

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by `createEmptyAnalysisSpecifications`

executionSettings An object of type ExecutionSettings as created by [createCdmExecutionSettings\(\)](#) or [createResultsExecutionSettings\(\)](#).

Method createResultsDataModel(): Create the results data model for the module

Usage:

```
SelfControlledCaseSeriesModule$createResultsDataModel(
  resultsConnectionDetails,
  resultsDatabaseSchema,
  tablePrefix = ""
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsDatabaseSchema The schema in the results database that holds the results data model.

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method getResultsDataModelSpecification(): Get the results data model specification for the module

Usage:

```
SelfControlledCaseSeriesModule$getResultsDataModelSpecification(
  tablePrefix = ""
)
```

Arguments:

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method uploadResults(): Upload the results for the module

Usage:

```
SelfControlledCaseSeriesModule$uploadResults(
  resultsConnectionDetails,
  analysisSpecifications,
  resultsDataModelSettings
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)

resultsDataModelSettings The results data model settings as created using [@seealso [createResultsDataModel](#)

Method createModuleSpecifications(): Creates the SelfControlledCaseSeries Module Specifications

Usage:

```
SelfControlledCaseSeriesModule$createModuleSpecifications(
  sccsAnalysesSpecifications,
  sccsAnalysisList = NULL,
  exposuresOutcomeList = NULL,
  analysesToExclude = NULL,
  combineDataFetchAcrossOutcomes = NULL,
  sccsDiagnosticThresholds = NULL
)
```

Arguments:

sccsAnalysesSpecifications An R6 class created by SelfControlledCaseSeries::createSccsAnalysesSpecification
sccsAnalysisList Deprecated with SelfControlledCaseSeries v6 - please use the sccsAnalysesSpecifications parameter instead.
exposuresOutcomeList Deprecated with SelfControlledCaseSeries v6 - please use the sccsAnalysesSpecifications parameter instead.
analysesToExclude Deprecated with SelfControlledCaseSeries v6 - please use the sccsAnalysesSpecifications parameter instead.
combineDataFetchAcrossOutcomes Deprecated with SelfControlledCaseSeries v6 - please use the sccsAnalysesSpecifications parameter instead.
sccsDiagnosticThresholds Deprecated with SelfControlledCaseSeries v6 - please use the sccsAnalysesSpecifications parameter instead.

Method validateModuleSpecifications(): Validate the module specifications

Usage:

```
SelfControlledCaseSeriesModule$validateModuleSpecifications(
  moduleSpecifications
)
```

Arguments:

moduleSpecifications The SelfControlledCaseSeries module specifications

Method clone(): The objects of this class are cloneable with this method.

Usage:

```
SelfControlledCaseSeriesModule$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

StrategusModule

StrategusModule defines the base class for each HADES module

Description

StrategusModule serves as an internal base class that defines the core functions and structure to be inherited and implemented by any specific HADES module. It provides a standardized framework for creating modular components within the Strategus pipeline.

Public fields

`moduleName` The name of the module taken from the class name. This is set in the constructor of the class.

`moduleClassName` The class name that identifies the module specifications in the overall analysis specification. This is set in the constructor of the class.

`internalModuleSpecificationClassName` A constant value. The base class name that identifies a module specification in the analysis specification.

`internalSharedResourcesClassName` A constant value. The class name that identifies the shared resources section in the overall analysis specification.

Methods

Public methods:

- [StrategusModule\\$new\(\)](#)
- [StrategusModule\\$execute\(\)](#)
- [StrategusModule\\$createResultsDataModel\(\)](#)
- [StrategusModule\\$getResultsDataModelSpecification\(\)](#)
- [StrategusModule\\$uploadResults\(\)](#)
- [StrategusModule\\$createModuleSpecifications\(\)](#)
- [StrategusModule\\$createSharedResourcesSpecifications\(\)](#)
- [StrategusModule\\$validateModuleSpecifications\(\)](#)
- [StrategusModule\\$validateSharedResourcesSpecifications\(\)](#)
- [StrategusModule\\$clone\(\)](#)

Method `new()`: Initialize the module

Usage:

```
StrategusModule$new()
```

Method `execute()`: Executes the module

Usage:

```
StrategusModule$execute(  
    connectionDetails,  
    analysisSpecifications,  
    executionSettings  
)
```

Arguments:

`connectionDetails` An object of class `connectionDetails` as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpecifications\(\)](#)

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by [createEmptyAnalysisSpecifications\(\)](#)

`executionSettings` An object of type `ExecutionSettings` as created by [createCdmExecutionSettings\(\)](#) or [createResultsExecutionSettings\(\)](#).

Method `createResultsDataModel()`: Create the results data model for the module

Usage:

```

StrategusModule$createResultsDataModel(
  resultsConnectionDetails,
  resultsDatabaseSchema,
  tablePrefix = ""
)

```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class `connectionDetails` as created by the `DatabaseConnector::createConnectionDetails()` function.

resultsConnectionDetails The connection details to the results database which is an object of class `connectionDetails` as created by the `DatabaseConnector::createConnectionDetails()` function.

resultsDatabaseSchema The schema in the results database that holds the results data model.

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method `getResultsDataModelSpecification()`: Get the results data model specification for the module

Usage:

```
StrategusModule$getResultsDataModelSpecification(tablePrefix = "")
```

Arguments:

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method `uploadResults()`: Upload the results for the module

Usage:

```

StrategusModule$uploadResults(
  resultsConnectionDetails,
  analysisSpecifications,
  resultsDataModelSettings
)

```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class `connectionDetails` as created by the `DatabaseConnector::createConnectionDetails()` function.

resultsConnectionDetails The connection details to the results database which is an object of class `connectionDetails` as created by the `DatabaseConnector::createConnectionDetails()` function.

analysisSpecifications An object of type `AnalysisSpecifications` as created by `createEmptyAnalysisSpec`

analysisSpecifications An object of type `AnalysisSpecifications` as created by `createEmptyAnalysisSpec`

resultsDataModelSettings The results data model settings as created using [@seealso `createResultsDataModel`

Method `createModuleSpecifications()`: Base function for creating the module settings object. Each module will have its own implementation and this base class method will be used to ensure the class of the specifications is set properly.

Usage:

```
StrategusModule$createModuleSpecifications(moduleSpecifications)
```

Arguments:

moduleSpecifications An object of type ModuleSpecifications

moduleSpecifications An object of type ModuleSpecifications

Method createSharedResourcesSpecifications(): Base function for creating the shared resources settings object. Each module will have its own implementation if it needs to create a shared resource.

Usage:

```
StrategusModule$createSharedResourcesSpecifications(
  className,
  sharedResourcesSpecifications
)
```

Arguments:

className The class name of the shared resources specifications

sharedResourcesSpecifications The shared resources specifications

Method validateModuleSpecifications(): Base function for validating the module settings object. Each module will have its own implementation and this base class method will be used to ensure the module specifications are valid ahead of execution

Usage:

```
StrategusModule$validateModuleSpecifications(moduleSpecifications)
```

Arguments:

moduleSpecifications An object of type ModuleSpecifications

moduleSpecifications An object of type ModuleSpecifications

Method validateSharedResourcesSpecifications(): Base function for validating the shared resources specification settings object. Each module will have its own implementation and this base class method will be used to ensure the module specifications are valid ahead of execution

Usage:

```
StrategusModule$validateSharedResourcesSpecifications(
  className,
  sharedResourcesSpecifications
)
```

Arguments:

className The class name of the shared resources specifications

sharedResourcesSpecifications The shared resources specifications

Method clone(): The objects of this class are cloneable with this method.

Usage:

```
StrategusModule$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

TreatmentPatternsModule

Evaluate phenotypes with the R <https://github.com/darwin-eu/TreatmentPatterns/DARWIN> *TreatmentPatterns Package*

Description

Characterization and description of patterns of events (cohorts). against the OMOP Common Data Model.

Super class

`Strategus::StrategusModule` -> TreatmentPatternsModule

Public fields

`tablePrefix` The table prefix to append to the results tables

Methods

Public methods:

- `TreatmentPatternsModule$new()`
- `TreatmentPatternsModule$execute()`
- `TreatmentPatternsModule$createResultsDataModel()`
- `TreatmentPatternsModule$getResultsDataModelSpecification()`
- `TreatmentPatternsModule$uploadResults()`
- `TreatmentPatternsModule$createModuleSpecifications()`
- `TreatmentPatternsModule$validateModuleSpecifications()`
- `TreatmentPatternsModule$clone()`

Method `new()`: Initialize the module

Usage:

`TreatmentPatternsModule$new()`

Method `execute()`: Execute Treatment Patterns

Usage:

```
TreatmentPatternsModule$execute(
  connectionDetails,
  analysisSpecifications,
  executionSettings
)
```

Arguments:

`connectionDetails` An object of class `connectionDetails` as created by the `DatabaseConnector::createConnectionDetails()` function.

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by `createEmptyAnalysisSpecifications()`

`analysisSpecifications` An object of type `AnalysisSpecifications` as created by `createEmptyAnalysisSpecifications()`

`executionSettings` An object of type `ExecutionSettings` as created by `createCdmExecutionSettings()` or `createResultsExecutionSettings()`.

Method createResultsDataModel(): Create the results data model for the module

Usage:

```
TreatmentPatternsModule$createResultsDataModel(
  resultsConnectionDetails,
  resultsDatabaseSchema,
  tablePrefix = self$tablePrefix
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsDatabaseSchema The schema in the results database that holds the results data model.

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method getResultsDataModelSpecification(): Get the results data model specification for the module

Usage:

```
TreatmentPatternsModule$getResultsDataModelSpecification(tablePrefix = "")
```

Arguments:

tablePrefix A prefix to apply to the database table names (optional).

tablePrefix A prefix to apply to the database table names (optional).

Method uploadResults(): Upload the results for TreatmentPatterns

Usage:

```
TreatmentPatternsModule$uploadResults(
  resultsConnectionDetails,
  analysisSpecifications,
  resultsDataModelSettings
)
```

Arguments:

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

resultsConnectionDetails The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)

analysisSpecifications An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpec](#)

resultsDataModelSettings The results data model settings as created using [@seealso [createResultsDataModel](#)

Method createModuleSpecifications(): Creates the TreatmentPatternsModule Specifications

Usage:

```
TreatmentPatternsModule$createModuleSpecifications(
  cohorts,
  includeTreatments = NULL,
  indexDateOffset = NULL,
  minEraDuration = 0,
  splitEventCohorts = NULL,
  splitTime = NULL,
  eraCollapseSize = 30,
  combinationWindow = 30,
  minPostCombinationDuration = 30,
  filterTreatments = "First",
  maxPathLength = 5,
  ageWindow = 5,
  minCellCount = 1,
  censorType = "minCellCount",
  overlapMethod = "truncate",
  concatTargets = TRUE,
  startAnchor = "startDate",
  windowStart = 0,
  endAnchor = "endDate",
  windowEnd = 0
)
```

Arguments:

cohorts (data.frame())

Data frame containing the following columns and data types:

cohortId numeric(1) Cohort ID's of the cohorts to be used in the cohort table.

cohortName character(1) Cohort names of the cohorts to be used in the cohort table.

type character(1) ["**target**", "**event**", "**exit**"] Cohort type, describing if the cohort is a target, event, or exit cohort

includeTreatments (character(1): "startDate")

DEPRECATED

"startDate" Include treatments after the target cohort start date and onwards.

"endDate" Include treatments before target cohort end date and before.

indexDateOffset (integer(1): 0)

DEPRECATED Offset the index date of the Target cohort.

minEraDuration (integer(1): 0)

Minimum time an event era should last to be included in analysis

splitEventCohorts (character(n): "")

Specify event cohort to split in acute (< X days) and therapy (>= X days)

splitTime (integer(1): 30)

Specify number of days (X) at which each of the split event cohorts should be split in acute and therapy

eraCollapseSize (integer(1): 30)

Window of time between which two eras of the same event cohort are collapsed into one era

combinationWindow (integer(1): 30)

Window of time two event cohorts need to overlap to be considered a combination treatment

minPostCombinationDuration (integer(1): 30)

Minimum time an event era before or after a generated combination treatment should last to be included in analysis

`filterTreatments` (character(1): "First" ["first", "Changes", "all"])
 Select first occurrence of ('First'); changes between ('Changes'); or all event cohorts ('All').

`maxPathLength` (integer(1): 5)
 Maximum number of steps included in treatment pathway

`ageWindow` (integer(n): 10)
 Number of years to bin age groups into. It may also be a vector of integers. I.e. `c(0, 18, 150)` which will results in age group 0-18 which includes subjects < 19. And age group 18-150 which includes subjects > 18.

`minCellCount` (integer(1): 5)
 Minimum count required per pathway. Censors data below x as <x. This minimum value will carry over to the sankey diagram and sunburst plot.

`censorType` (character(1))

"minCellCount" Censors pathways <minCellCount to minCellCount.
 "remove" Censors pathways <minCellCount by removing them completely.
 "mean" Censors pathways <minCellCount to the mean of all frequencies below minCellCount

`overlapMethod` (character(1): "truncate") Method to decide how to deal with overlap that is not significant enough for combination. "keep" will keep the dates as is. "truncate" truncates the first occurring event to the start date of the next event.

`concatTargets` (logical(1): TRUE) Should multiple target cohorts for the same person be concatenated or not?

`startAnchor` (character(1): "startDate") Start date anchor. One of: "startDate", "endDate"

`windowStart` (numeric(1): 0) Offset for startAnchor in days.

`endAnchor` (character(1): "endDate") End date anchor. One of: "startDate", "endDate"

`windowEnd` (numeric(1): 0) Offset for endAnchor in days.

Method `validateModuleSpecifications()`: Validate the module specifications

Usage:

`TreatmentPatternsModule$validateModuleSpecifications(moduleSpecifications)`

Arguments:

`moduleSpecifications` The CohortMethod module specifications

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

`TreatmentPatternsModule$clone(deep = FALSE)`

Arguments:

`deep` Whether to make a deep clone.

uploadResults

Upload results

Description

Upload the results for a given analysis

Usage

```
uploadResults(  
  analysisSpecifications,  
  resultsDataModelSettings,  
  resultsConnectionDetails  
)
```

Arguments

- analysisSpecifications
An object of type AnalysisSpecifications as created by [createEmptyAnalysisSpecifications\(\)](#)
- resultsDataModelSettings
The results data model settings as created using [[@seealso createResultsDataModelSettings\(\)](#)]
- resultsConnectionDetails
The connection details to the results database which is an object of class connectionDetails as created by the [DatabaseConnector::createConnectionDetails\(\)](#) function.

zipResults	Create a zip file with all study results for sharing with study coordinator
------------	---

Description

Create a zip file with all study results for sharing with study coordinator

Usage

```
zipResults(resultsFolder, zipFile)
```

Arguments

- resultsFolder The root folder holding the study results.
- zipFile The path to the zip file to be created.

Details

Creates a .zip file of the .csv files found in the resultsFolder. The resulting .zip file will have relative paths to the root of the resultsFolder which is generally found in executionSettings\$resultsFolder.

Value

Does not return anything. Is called for the side-effect of creating the zip file with results.

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